

2(a)	any four from: <table><tr><th>structure</th><th>function</th></tr><tr><td>1 thylakoid(s) / granum / grana</td><td>light dependent reaction / photophosphorylation ;</td></tr><tr><td>2 thylakoid <u>membrane</u>(s)</td><td>have (named), pigments / photosystems / LHC / electron carriers / ETC / ATP synthase ;</td></tr><tr><td>3 thylakoid membranes / grana, have large surface area / are many in number</td><td>to absorb (more) light ;</td></tr><tr><td>4 stroma</td><td>for light independent reaction / Calvin cycle ;</td></tr><tr><td>5 stroma</td><td>has, (named) enzymes / RuBP / reduced NADP ;</td></tr><tr><td>6 stroma colourless OR stroma contains water</td><td>so light reaches thylakoids OR as a medium for reactions / for photolysis</td></tr><tr><td>7 DNA / ribosomes</td><td>make (named), chloroplast proteins / proteins for photosynthesis ;</td></tr><tr><td>8 starch grains / lipid droplets</td><td>store, (named) product of photosynthesis / (chemical) energy ;</td></tr><tr><td>9 envelope</td><td>compartmentalisation ;</td></tr></table>	structure	function	1 thylakoid(s) / granum / grana	light dependent reaction / photophosphorylation ;	2 thylakoid <u>membrane</u> (s)	have (named), pigments / photosystems / LHC / electron carriers / ETC / ATP synthase ;	3 thylakoid membranes / grana, have large surface area / are many in number	to absorb (more) light ;	4 stroma	for light independent reaction / Calvin cycle ;	5 stroma	has, (named) enzymes / RuBP / reduced NADP ;	6 stroma colourless OR stroma contains water	so light reaches thylakoids OR as a medium for reactions / for photolysis	7 DNA / ribosomes	make (named), chloroplast proteins / proteins for photosynthesis ;	8 starch grains / lipid droplets	store, (named) product of photosynthesis / (chemical) energy ;	9 envelope	compartmentalisation ;	4
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2(b)	any three from: 1 light excites electrons in, P700 / PS1 (reaction centre chlorophyll) ; 2 electrons passed, along carriers / to ETC ; 3 H⁺ / protons, pumped into <u>thylakoid</u>, space / lumen ; 4 H⁺ / protons, diffuse (back to stroma) through ATP synth(et)ase ; 5 makes ATP ; 6 ADP ;	3																				

Question	Answer	Marks
2(c)	<i>any seven from:</i> <i>description</i> D1 identify wavelengths for chlorophyll (a) absorption ; D2 identify wavelengths for phycoerythrin absorption ; D3 identify wavelengths for phycocyanin absorption ; D4 chlorophyll (a), has peaks / absorbs mainly, in blue and red ; D5 phycoerythrin absorbs in blue and green and yellow ; D6 phycocyanin absorbs in (green) yellow and red ; <i>explanation</i> E7 red algae / deep water, get(s) green (and yellow) light ; E8 chlorophyll (a) absorbs, no / little, green (and yellow) light ; E9 phycoerythrin / phycocyanin / accessory pigments, absorbs wavelengths not absorbed by chlorophyll (a) ; E10 combined pigments absorb, greater range of / any / all, wavelengths ; E11 increases / more / high(er) rate of, light dependent stage / photosynthesis ; E12 (so) more / lots of, (named) organic compounds / growth ;	7

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